

BeliefLens: Belief Dynamics in LLM Forecasting Harnesses

Technical Report

BeliefLens Project

Deep Digital Intelligence · <https://github.com/DDI-droid/belieflens>

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Abstract

LLM forecasters are *judgmental* forecasters: behind every stated probability is something that behaves like a belief, and behind every revision an update rule. BeliefLens measures both. Four reasoning harnesses (**react**, **futuresim**, **analytica**, **bayesian**), built as prompts over one shared substrate on a 339,801-article date-gated news corpus, forecast real January–March 2026 events under two protocols: **sequential** (five dates in order, own history carried; $k=5$ rollouts) and **parallel** (fresh one-shot per date; $k_2=5$ rollouts), so every belief is a distribution. From each trajectory we *recover the reasoning as a literal-free program*: named variables extracted from the reasoning are the only values a synthesized program may use, and blind sample/temporal holdouts decide whether the recovered method is real. Findings: reasoning-structure stability and evidence tracking *anticorrelate*; **bayesian**’s reasoning is a genuine fixed formula (its early-days program predicts held-out late days at $\text{MAE} \approx 0.003$) and its sequential trajectory is unbiased against its own fresh-eyes belief; **futuresim**’s rigidity decomposes into history anchoring (update gain -0.14 , the only negative), procedure-level conservatism, and—under direct evidence—full rule compliance; **react**’s updates flow through channels no recoverable program contains. All 1,432 logged searches replay with zero articles past the date gate, and an independent adversarial review is incorporated with every correction documented (§8).

1 Introduction

Benchmarks such as FutureSim [2] score LLM forecasting agents on outcomes, and observe behaviours—anchoring to initial forecasts, “self-conditioning” on past rationales—that are claims about *belief dynamics*. But a score cannot say where a pathology lives: in what the agent *believes*, in how it *updates*, or in how it *gathers evidence*. This report develops instrumentation that separates those layers, and applies it to four widely used reasoning shapes.

The central device is **program recovery** (§6). If a harness’s day-to-day reasoning is a stable method applied to changing inputs, then a single program—with *no numeric literals anywhere*, only named quantities extracted from the reasoning itself—should reproduce every forecast the harness made, with only the variable bindings changing between days. The program is then the *reasoning structure*; the bindings are the *belief state*; and the split is forced by construction, because the literal ban leaves a number nowhere to hide. Once such a program exists, belief dynamics become measurable: the belief is a trajectory in the program’s own coordinates, and the program is an executable referee for counterfactual probes (§7).

Contributions.

- **Infrastructure.** A replayable forecasting environment on the real FutureSim corpus (339,801 articles, 2025-12-01–2026-03-31) with an SQL-enforced date gate verified by *replaying all 1,432 logged searches* (zero breaches); four harness shapes on one substrate so that reasoning structure is the only manipulated variable; dual elicitation protocols (sequential $k=5$, parallel $k_2=5$) yielding belief *distributions*, not points.

harness	structure imposed	carries across days	lineage
react	free think/search loop	forecast history*	ReAct [4]
futuresim	6 ordered guidelines; base rate first; forced ≤ 200 -word memory update	history + memory block	FutureSim baseline harness [2]
analytica	decompose 3–5 sub-propositions $\rightarrow$ ground each by search $\rightarrow$ synthesize with weights	forecast history*	Analytica [1]
bayesian	prior $\rightarrow$ evidence $\rightarrow$ per-item likelihood ratio $\rightarrow$ posterior; forced belief block	history + belief block	sequential Bayesian updating [5]

Table 1: The four reasoning shapes. *All four see their own prior forecasts in the sequential protocol (the anchoring pressure, applied uniformly); **react** is therefore a *history-anchored* floor, not a memoryless one.

- **Method.** A literal-free program space with a mechanical anti-gaming validator (no literals, no boolean constants, no constant-only arithmetic, no value-spelling names such as `two=2`, no dead code, no x/x), an answer-copy filter over raw/odds/percent/complement transforms, and blind sample/temporal holdouts as the deciding gate; a standing audit that re-verifies all of it against collected data.
- **Findings.** A stability–plasticity inversion across harnesses; recovery of a true fixed update formula for **bayesian**; a three-way decomposition of **futuresim**’s rigidity (acquisition, procedure, and history—but *not* rule violation); quantified conservatism (update gains 0.22–0.47, all < 1) with **futuresim** negative; and the incompressibility of free-form ReAct reasoning.

2 Background

Analytica [1] decomposes a query into a tree of soft propositions, grounds leaves with tools, and composes upward with a linear rule, arguing bias falls at the leaves and variance in the composition. **FutureSim** [2] replays dated news past model cutoffs and documents anchoring and self-conditioning in frontier agents. **Agent-BRACE** [3] represents an agent’s belief as atomic claims with ordinal certainty and shows such beliefs grow calibrated as evidence accrues. Our harnesses instantiate these three shapes (plus a plain ReAct [4] floor); our measurements ask the question none of the three answers: whether the externalized structure *is* the working belief, and what moves it.

3 Experimental Infrastructure

3.1 Corpus and date gate

Articles are the FutureSim corpus (HF `shash42/forecast-news`; CCNews-derived, deduplicated, dated): 120 day-partitions, 2025-12-01–2026-03-31, indexed to SQLite FTS5 (BM25), 339,801 rows. At simulation date D the one search tool returns only articles with date $\leq D$; the constraint is in the SQL, not the prompt. *Verification:* the audit re-executes every search any harness ever logged (483 in the main runs, 949 in the scale-up) with its turn’s date; zero returned articles postdate the gate. We do not use FutureSim’s 48 GB prebuilt hybrid index; retrieval (BM25) is weaker than the paper’s but byte-identical across harnesses, so it is a constant of the experiment, not a confound in it.

3.2 Harnesses

Four system prompts over one substrate (same model, tools, tool budget, output contract `FORECAST: p`); Table 1. Installing the papers’ repositories would compare repositories; holding the substrate fixed makes reasoning shape the only variable.

3.3 Model choice and contamination

Harnesses run on `gpt-5-mini` (released 2025-08-07): the questions resolve January–March 2026, so any later model would *recall* rather than forecast, and a model that knows the outcome does not move its belief—

destroying the dynamics measurement. The extractor (gpt-5.2) post-dates the events; it writes variables and programs but makes no forecasts, and its one leakage channel (tilting extracted values toward the known outcome) is audited via the preserved pre-reconciliation variable tables: across 320 tables, exactly one systematically answer-tracking variable was found (`posterior_odds`, odds-form, 10/10 instances), and no recovered program references it.

3.4 Questions

The paired control: *Will the USA / will Canada win the Milano-Cortina men’s hockey gold?* (one evidence stream; resolves YES/NO). Both teams won their semifinals on 02-20—the last simulated date—and the final resolved 02-22, *outside* the window; a correct forecaster therefore raises *both* beliefs toward ~ 0.5 , and the proper metrics are coherence metrics—both beliefs rising across the semifinal (*semi-rise*) and $P_{\text{USA}} + P_{\text{CAN}} \rightarrow 1$ —not directional separation. (Our original design scored directional separation; the adversarial review, §8, showed it would have punished correct forecasting, and it was replaced before any conclusion was drawn.) A scale-up set of four further questions from the OpenForesight dataset (FutureSim’s question source) supports the recovery statistics of §6; all four resolve YES, a bias we disclose and therefore exclude from directional claims.

4 The Literal-Free Program Space

A recovered program is one Python function `forecast()` over an injected namespace, with: **(R1)** no numeric, string, or *boolean* literals anywhere; **(R2)** no user-defined functions, imports, attributes, subscripts, comprehensions, or containers (recursion and `while` permitted; execution carries a 200k-step budget so non-termination is a failed fit); **(R3)** calls only to eleven whitelisted operations (`clamp`, `logistic`, `noisy_or`, `odds`, `prob`, `min/max/abs/round/ exp/log/sqrt`) and nine named constants; **(R4)** *no value may hide in a name*: `two=2`, `half=0.5`, `const_0_62` are rejected in variable tables, schemas, and program locals alike; **(R5)** no constant-only arithmetic (`TENTH*(TWO+TWO+TWO)` = a smuggled 0.6), no self-ops ($x-x$, x/x), no dead code (`ZERO*(...)`), identity ops, unused locals). An eleven-exploit battery (including all constructions found by the adversarial review) must reject under the validator and does; a legitimate program passes. Because in-sample fit with visible targets cannot by itself distinguish structure from memorisation, **the deciding gate is out-of-sample**: programs are re-synthesized on a subset whose complement’s targets the synthesizer never sees.

5 Experiment 1: Replay

5.1 Protocols

Sequential (seq): five dates in order (01-14, 01-28, 02-08, 02-15, 02-20); each day the harness sees its own previous forecasts (and its carry block) and may search the gated corpus; $k=5$ independent rollouts of the whole trajectory. **Parallel (par)**: for each date, a fresh single-shot run given all information up to that date and *no* history; $k_2=5$ rollouts per date. Day 1 of a sequential rollout is, by construction, a parallel draw—a built-in sanity check (12/16 day-1 sequential values fall inside the day-1 parallel spread at ± 0.05). All 560 collected forecasts parse under the strict parser (200 sequential + 200 parallel on the pair, plus 160 scale-up; no fallback; percent and comma handled; out-of-range triggers a retry), and every stored forecast re-parses identically from its own turn text. Figure 1 shows the resulting trajectories.

5.2 Metrics

For dates $t \geq 2$: **anchor gap** = $\text{SEQ}_t - \overline{\text{PAR}}_t$ (signed; what carrying history does relative to evidence alone); **lag advantage** = $|\text{SEQ}_t - \overline{\text{PAR}}_t| - |\text{SEQ}_t - \overline{\text{PAR}}_{t-1}|$ (positive $\Rightarrow$ the sequence tracks *yesterday’s* evidence-conditioned belief better than today’s); **update gain** = OLS slope of ΔSEQ_t on $\Delta \overline{\text{PAR}}_t$ (1 = moves with the evidence; < 1 conservatism); **placement** = percentile of each SEQ forecast within that date’s PAR draws (50 = unbiased); **stabilisation** = $\text{sd}(\text{SEQ})/\text{sd}(\text{PAR})$ per date; **coherence** = $P_{\text{USA}} + P_{\text{CAN}}$ (**sum@last** is its value

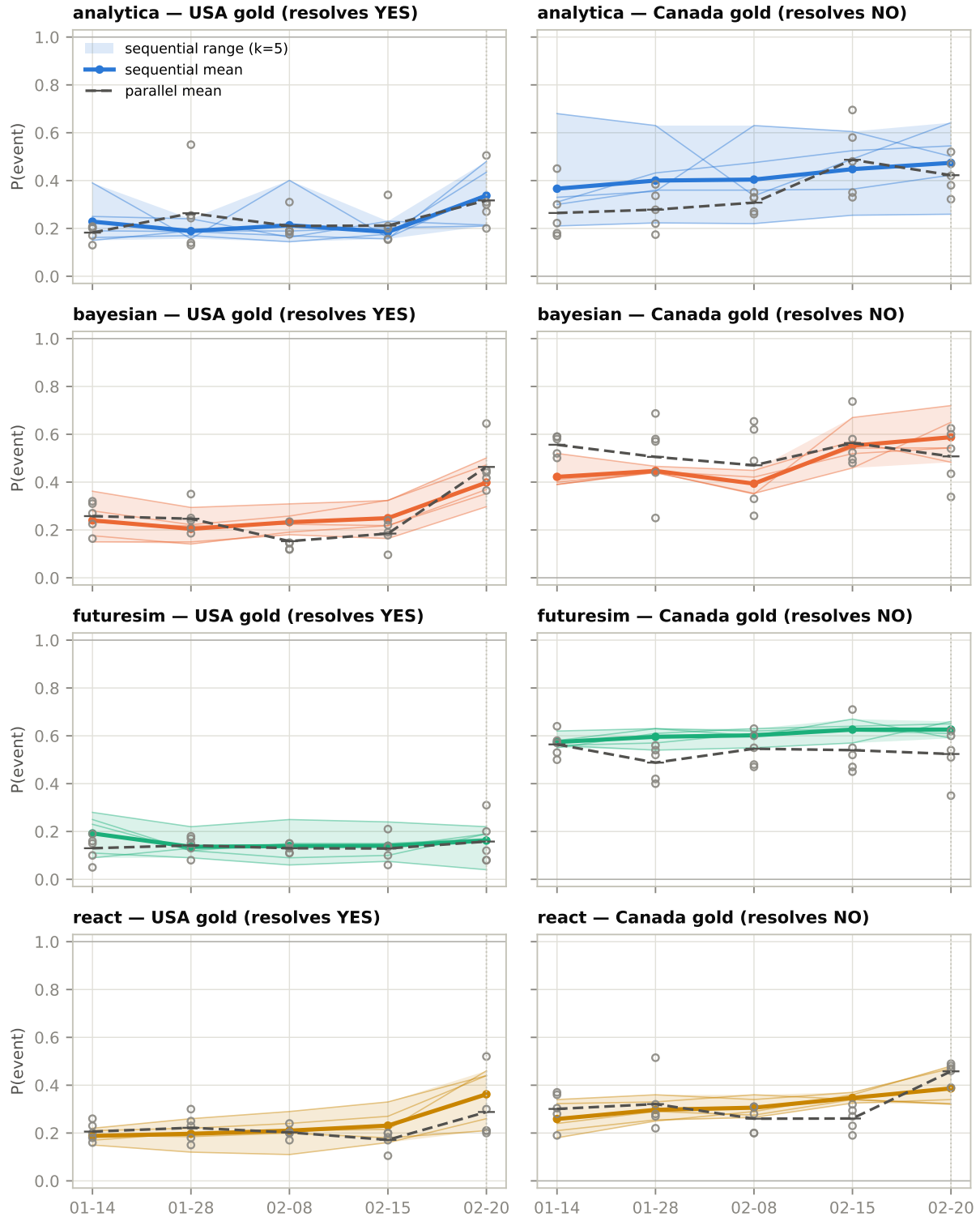


Figure 1: Belief flow, all four harnesses on the paired questions. Shaded band and bold line: range and mean of the five SEQ rollouts (history carried). Open dots and dashed line: the five PAR one-shot draws per date and their mean (fresh eyes on the same evidence). Dotted vertical: semifinal date. **futuresim**–Canada’s band rides above its fresh-eyes mean (anchoring) and **futuresim**–USA barely responds even in PAR (procedure-level conservatism); **bayesian** hugs its fresh-eyes mean in both panels.

harness	replay (SEQ, $k=5$)			SEQ vs PAR				
	$ \Delta _{\text{tot}}$	semi-rise	sum@last	gap	gap	gain	lag adv.	PIT _{med}
analytica	0.331	0.60	0.81	+0.019	0.098	0.22	−0.009	60
bayesian	0.283	0.80	0.99	− 0.004	0.072	0.47	−0.028	40
futuresim	0.134	0.00	0.79	+ 0.046	0.067	− 0.14	+ 0.003	80
react	0.182	0.60	0.75	+0.019	0.066	0.27	−0.011	60

Table 2: E1 headline metrics ($k=5$ both conditions; pooled over the pair). Semi-rise: fraction of rollouts in which *both* paired beliefs rose across the semifinal date (the correct direction). **bayesian** is the best coherent tracker and is unbiased against its own fresh-eyes belief; **futuresim** never raised both after the semifinals, carries the largest anchor gap, is the only harness with positive lag advantage, and its update gain is *negative*—together consistent with tracking yesterday’s evidence-conditioned belief rather than today’s. Bold marks, per column, the value most indicative of anchoring or of its absence.

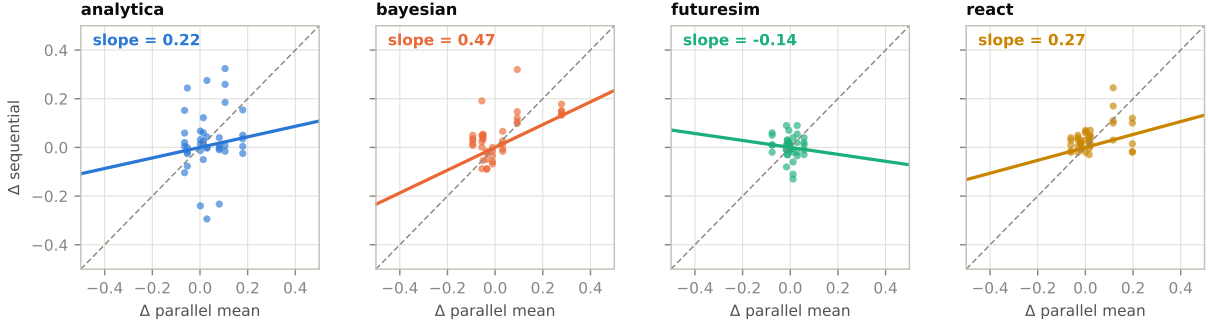


Figure 2: Update gain. Each dot is one date-step: x , movement of the fresh-eyes mean; y , movement of a sequential rollout. Dashed: $y=x$. All fitted slopes are below 1 (conservatism); **futuresim**’s is negative.

on 02-20); **total movement** $|\Delta|_{\text{tot}}$, the summed absolute day-to-day change of a rollout; and **Brier** [6]. Placement is a probability-integral-transform (PIT) statistic, and PIT_{med} its median. Figures 2–5 render these; Table 2 summarises. Throughout this report, Brier is the plain quadratic score $\frac{1}{n} \sum_i (p_i - y_i)^2$ with $y_i \in \{0, 1\}$ the resolved outcome, pooled over the pair—not a skill score (no baseline subtraction, no 1–MSE orientation, no difficulty adjustment); 0 is perfect, 0.25 is the constant-0.5 forecaster.

5.3 Search effort

Mean searches per turn (SEQ / PAR): **analytica** 7.3/6.8, **futuresim** 6.6/6.2, **react** 5.3/5.0, **bayesian** 5.0/5.5. Memory does not substitute for retrieval at this scale; **analytica**’s per-sub-proposition grounding makes it the heaviest searcher in both conditions (a truncation covariate logged per turn as **forced_stop**).

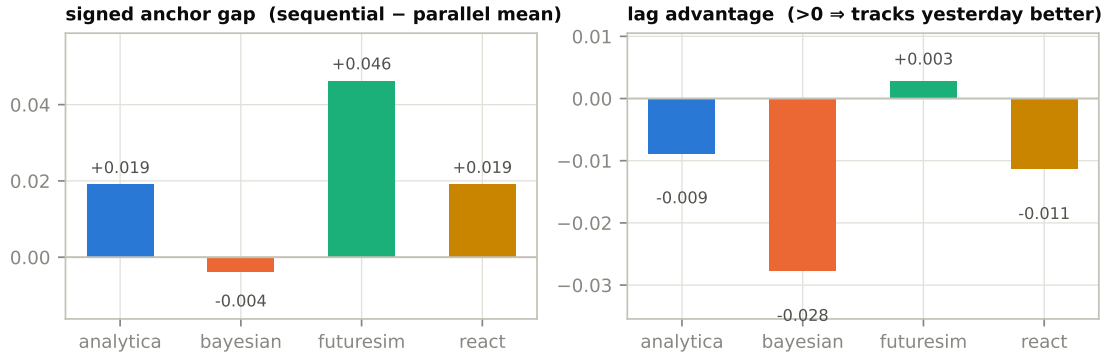


Figure 3: Anchoring summary. Left: signed anchor gap. Right: lag advantage; only *futuresim* is positive.

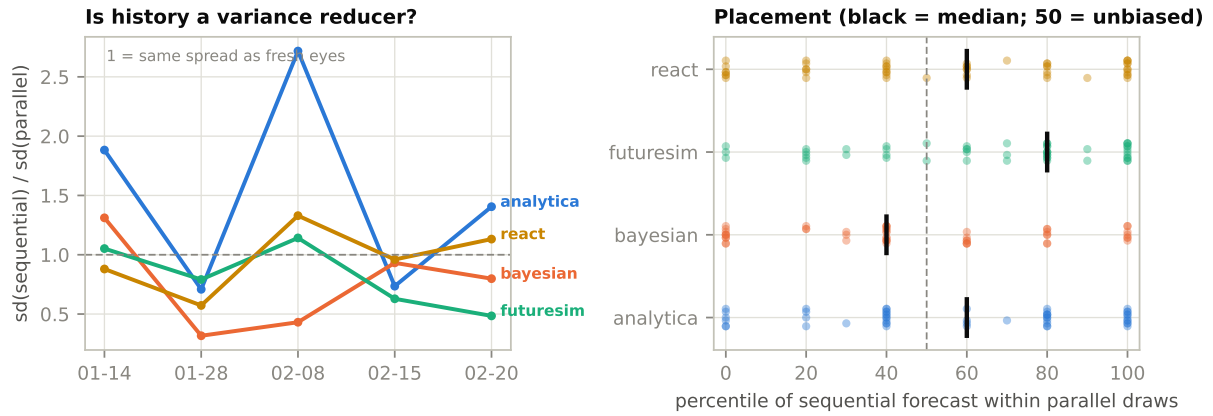


Figure 4: Left: stabilisation ratio $sd(SEQ)/sd(PAR)$. History acts as a variance reducer where the ratio sits below 1 (*futuresim*, *react*); *analytica*'s ratio above 1 on several dates is path dependence—rollouts diverge because their own early draws differed. **Right:** placement of every sequential forecast within its date's fresh-eyes distribution; *futuresim*'s median sits at the 80th percentile (anchored high), *bayesian*'s at the 40th.

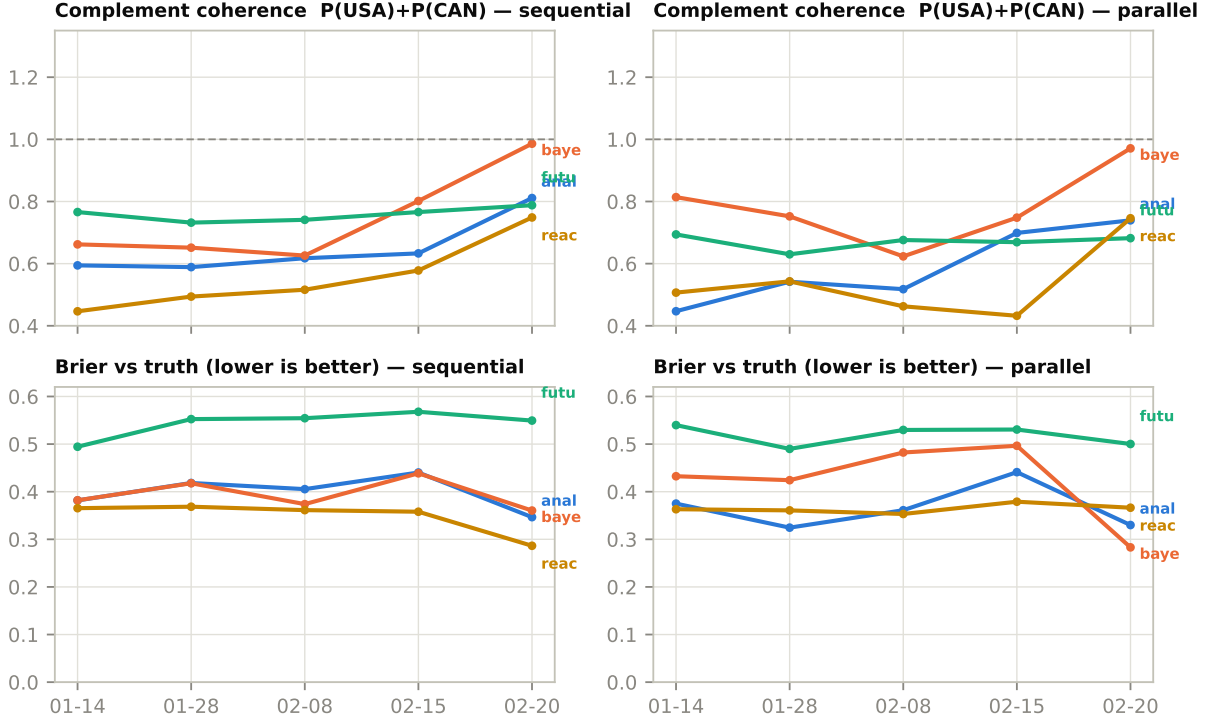


Figure 5: Coherence and accuracy by condition. Top: $P_{\text{USA}}+P_{\text{CAN}}$ should approach 1 by the semifinal (dashed). Bottom: plain Brier against the resolved outcomes (lower is better), pooled over the pair.

6 Experiment 2: Program Recovery

6.1 Pipeline

For every forecast turn: **step 1**, the extractor reads that day’s searches and reasoning and emits **name = value** declarations for every quantity the reasoning used—never the output, in any parameterisation (raw, odds, percent, complement all dropped within ± 0.005 ; yesterday’s forecast exempted as the legitimate anchor input); names must describe world quantities, never values. **Step 2** reconciles the per-day tables into one canonical schema (missing instance keys retried once, then excluded loudly—never zero-filled; pre-reconciliation tables preserved as the audit trail). **Step 3** synthesizes *one* program per harness-question against all (≤ 10) instances, repaired against validator violations and worst residuals for at most four attempts; then re-synthesizes *blind* on (a) sample 0 only and (b) the first three dates only, evaluated untouched on the held-out complement. Figure 6 and Table 3 give the results.

6.2 Residual drift and theory change

Fitting one program across all dates makes *failure* informative: *analytica*/CAN’s mean $|\text{residual}|$ grows by $+0.34$ from first to last date (USA: $+0.05$)—its method decays—while *futuresim*’s is flat or improving (CAN -0.007 , USA -0.07). Figure 7 shows the recovered belief coordinates for the cleanest case. Function-space distance between the full-fit and sample-0-only programs (Monte-Carlo over the observed binding box): *futuresim*/CAN **0.000** (identical theory from either half of the data), *bayesian*/USA 0.044, *react*/CAN 0.174, *analytica*/CAN 0.391.

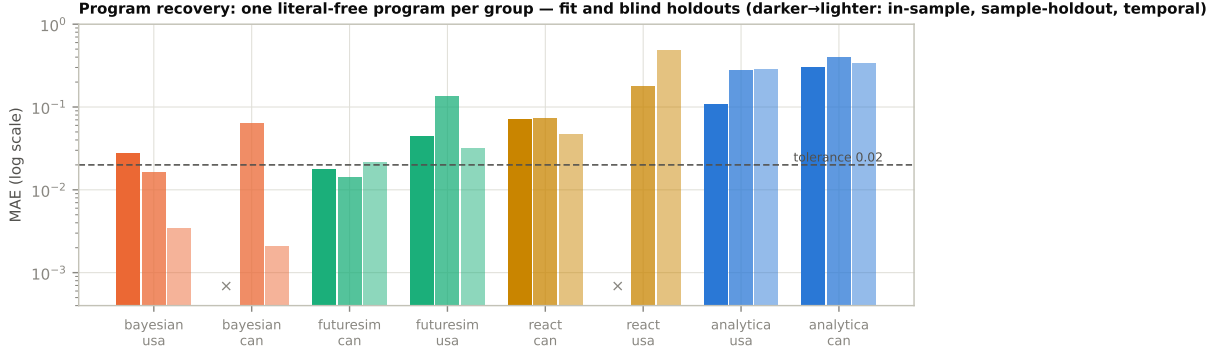


Figure 6: Recovery quality per group (log MAE; × marks “no valid program”). **bayesian**’s blind temporal holdouts reach MAE 0.002–0.003: a genuinely fixed update formula. **analytica** fits poorly under every regime—its structure changes day to day. **react**–USA admits no program at all.

group	in-sample	sample-holdout	temporal (early→late)	reading
bayesian/USA	0.027	0.016	0.0005 → 0.003	fixed log-odds formula
bayesian/CAN	—	0.063	0.0003 → 0.002	same; full-fit failed to execute
futuresim/CAN	0.018	0.014	0.015 → 0.021	stable; both halves recover the <i>identical</i> program
futuresim/USA	0.045	0.135	0.051 → 0.032	stable-ish
react/CAN	0.070	0.073	0.044 → 0.046	weak method
react/USA	none	0.178	0.252 → 0.487	incompressible
analytica/USA	0.108	0.274	0.073 → 0.285	structure changes daily
analytica/CAN	0.301	0.396	0.520 → 0.335	

Table 3: Program-recovery MAE. Holdout columns are blind (targets hidden from the synthesizer); bold marks MAE at or below the 0.02 tolerance. No group passes the strict gate *on the full in-sample fit* ($\text{MAE} \leq 0.02 \wedge \geq 80\%$ of instances within tolerance $\wedge 0$ failures), so in-sample fits are reported as descriptive—but two blind holdouts individually meet every threshold (**futuresim**/CAN sample-holdout: 0.014 at 80% in-tolerance, 0 failures; **bayesian** temporal: 0.002–0.003 at 100%), and the blind columns are the honest evidence throughout. Scale-up (four further questions, 16 groups, hardened rules): 10 clean, 5 marked recovery-uninformative (near-constant targets make both copy-detection and recovery ill-posed), 1 no-program.

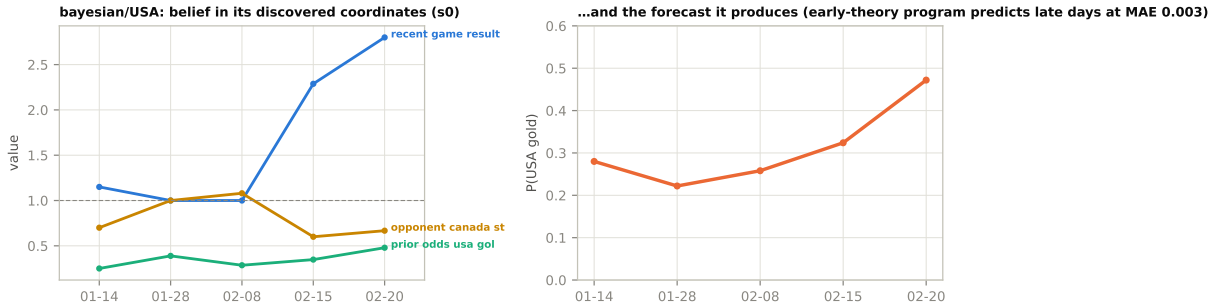


Figure 7: The belief in its discovered coordinates (**bayesian**/USA, rollout s0). Left: three recovered variables across the window—the recent-results likelihood ratio climbs $1.1 \rightarrow 2.9$ as the USA advances while the prior odds ratchet behind it. Right: the forecast those coordinates produce. The recovered programs are log-odds additive updating with a shrink-to-prior term (full-fit version listed in Appendix A); re-synthesized blind on the first three dates alone, the temporal variant—same family, different expression—predicts the last two days at the MAE shown.

7 The Rigidity Probe

Each probed harness was continued from its real end-of-run state to a probe day (02-21) with search disabled (stated as an outage) and exactly one injected, role-based news item—“Canada’s starting goaltender ruled out of the final” / the USA mirror / an irrelevant placebo. Its own recovered program supplies the referee twice over: a *single-coordinate* implied change (perturb only the targeted variable), and a *multi-coordinate* rule-implied forecast (re-extract the probe day’s full bindings from the harness’s own reasoning and execute the program). Placebo movements were ≤ 0.02 ; **all sixteen directional probes (4 groups $\times$ 2 rollouts $\times$ 2 directions) moved the correct direction**. Figure 8 and Table 4 summarise.

harness	method stable?	tracks evidence?	honours own rule?	diagnosis
futuresim	yes	poorly in replay	yes (gap ≤ 0.04)	rigidity lives in <i>acquisition</i> , not revision
bayesian	yes	good	asymmetric	compliant on good news; over-drops on bad news vs own rule (≈ -0.135)
react	no	yes	n/a (rule has no channel)	updates outside recoverable structure
analytica	no	good	no verdict possible	rewrites its method daily

Table 4: Per-harness verdicts, combining §5–§7. The one-line finding: structural stability and evidence-tracking anticorrelate, and the probe *localises* each harness’s rigidity or plasticity.

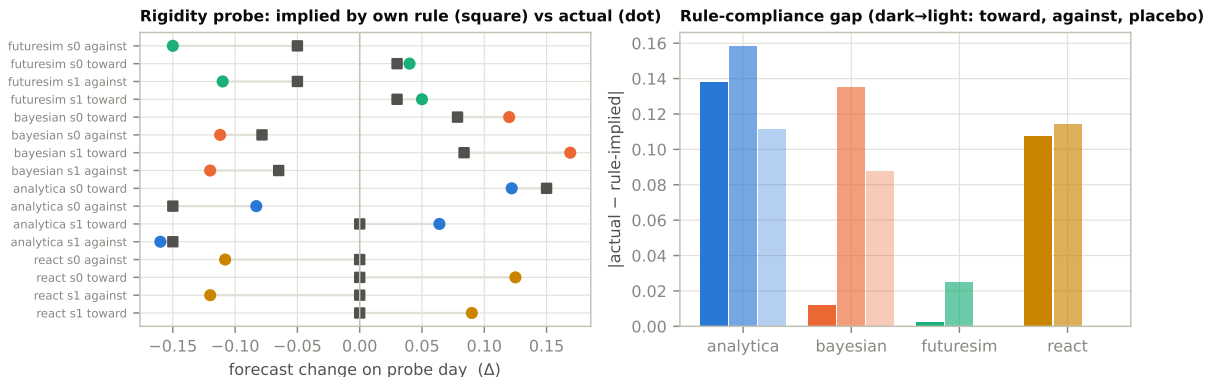


Figure 8: Left: single-coordinate implied change (square) vs actual change (dot) for every directional probe. **Right:** the multi-coordinate rule-compliance gap $|p_{\text{actual}} - p_{\text{rule}}|$ by condition. **futuresim**’s gaps are ≤ 0.04 : handed the evidence, it computes with its recovered structure almost exactly. **react**’s program implies zero response in every cell while the live harness moved ± 0.09 – 0.13 : its updates use machinery its recoverable structure does not contain.

8 Adversarial Review and Standing Audit

Mid-experiment, an independent reviewer was pointed at the entire codebase with instructions to break it. It returned 2 fatal, 7 serious, and 6 moderate findings; every fixable one was patched and re-verified before results were trusted. The fatals: (F1) the planned “separation” metric would have punished correct forecasting given the actual event path (§3); (F2) the no-literal rule was semantically bypassable (constant-construction, boolean arithmetic, if-chain lookup tables coached by the repair loop)—closed by the rules of §4 plus the blind holdouts. Serious findings included a thread-unsafe SQLite connection that corrupted 32% of an early run (that run is quarantined and unused), a forecast parser that could fabricate certainty from “85%” (fixed; the strict parser reproduces all collected forecasts exactly), and silent zero-filling in schema reconciliation (replaced by loud exclusion plus a preserved audit trail). The standing audit (`scripts/audit_exp1.py`) re-verifies on demand: 11/11 exploits rejected; all 560 collected forecasts parse under the strict parser (the 240 audited retroactively re-parse identically); 1,432 searches replayed with zero gate breaches; every stored program re-executes to its reported MAE; one systematic answer-copy in 320 variable tables, referenced by no program.

9 Limitations

(1) One question pair carries all directional claims; the scale-up set is all-YES by construction and supports only recovery statistics. (2) $k=5$ per cell bounds distributional claims; PIT placement has 5-draw granularity. (3) Draws are ordinary decoding variation (current reasoning endpoints reject temperature control). (4) Stated beliefs only: all measurements are consistency properties of verbalized probabilities under intervention, not claims about internal representations. (5) Retrieval is BM25, weaker than FutureSim’s hybrid index—held constant across harnesses, but absolute accuracy is a lower bound. (6) The extractor knows the outcomes; the audit bounds but does not eliminate this channel. (7) Program recovery inherits the expressiveness of the program space; “incompressible” means no program *in this space* within four repair attempts.

10 Conclusion and Future Work

Treating a forecasting agent’s reasoning as a recoverable, literal-free program turns belief dynamics from a metaphor into a measurement: beliefs become trajectories in discovered coordinates, updates become testable against the agent’s own rule, and rigidity decomposes into acquisition, procedure, history, and revision

components that this report measures separately for four harness shapes. Next: paired NO-questions and longer date grids; per-search-call budgets; a code-execution harness as the recovery upper bound; program recovery on the parallel condition (does removing history delete the anchor variables from the schema?); and alignment of the `bayesian` harness with its source paper’s exact protocol.

References

- [1] *Analytica: Soft Propositional Reasoning for Robust and Scalable LLM-Driven Analysis*. arXiv:2604.23072, 2026.
- [2] *FutureSim: Replaying World Events to Evaluate Adaptive Agents*. arXiv:2605.15188, 2026.
- [3] *Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty*. arXiv:2605.11436, 2026.
- [4] *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR, 2023.
- [5] *Agentic Forecasting using Sequential Bayesian Updating of Linguistic Beliefs*. arXiv:2604.18576, 2026.
- [6] *Verification of forecasts expressed in terms of probability*. Monthly Weather Review, 1950.

A A Recovered Program

The `bayesian`/USA program (full-fit). The blind variants recover the same log-odds-additive family, though not this exact expression. Note the absence of literals; every quantity is a step-1 variable or a named constant. The schema contained an odds-form copy of the answer (`posterior_odds_usa_gold`); the program does not reference it.

```
def forecast():
    p0 = prior_probability_usa_gold
    o0 = prior_odds_usa_gold
    ln_prior_odds = log(o0)
    ln_lr_total = log(lr_roster_strength_positive)
    ln_lr_total += log(lr_opponent_canada_strength)
    ln_lr_total += log(lr_injury_and_replacements)
    ln_lr_total += log(lr_recent_game_results)
    ln_lr_total += log(lr_nhl_participation_logistics)
    ln_lr_total += log(lr_venue_or_media_sentiment)
    ln_post_odds = ln_prior_odds + ln_lr_total
    p_bayes = logistic(ln_post_odds)
    logit_p0 = log(p0) - log(ONE - p0)
    gap = ln_post_odds - logit_p0
    tight = ONE / (ONE + abs(gap))
    w = (HALF + tight) * HALF
    return p0 + w * (p_bayes - p0)
```

B Reproduction

`git clone https://github.com/DDI-droid/belieflens` && `python bootstrap.py` downloads the corpus (~412 MB), builds the index, and runs the verification battery; `scripts/` contains every stage (`run_exp1.py` for SEQ/PAR/E2, `audit_exp1.py`, `beliefdyn.py`, `run_probe.py`, `probe_rule_check.py`, `make_charts.py`). All collected data, configs, and logs ship in `results/`.